

Technical Documentation

1. Problem framing

Forecast e-commerce **revenue** and **ROAS** as probabilistic ranges (P10 / P50 / P90) over 30 / 60 / 90-day windows, broken down by channel and campaign type, with support for future-budget simulation and AI-assisted explanation.

We deliberately frame this as an **aggregate-period, probabilistic** problem (per the brief): the business decision needs a credible *range* over a planning window, not a precise daily point estimate.

2. Data & preprocessing

Three platform exports (Google Ads, Microsoft/Bing Ads, Meta Ads), each with a different schema. They are normalized to one canonical schema via per-platform **adapters**:

Issue	Handling
Different column names	adapter renames to canonical schema
Google spend in micros	divided by 1,000,000 to dollars
Campaign-type spellings (PERFORMANCE_MAX vs PerformanceMax)	mapped to one label
Meta has no revenue / type	revenue left missing; Meta excluded from revenue model (treated spend-only)
Trailing partial month	dropped (filter_complete_months) — this was the single biggest accuracy fix
All-missing monthly sums	min_count=1 keeps them missing instead of a fake \$0

Platform detection is by **column signature**, not filename, so renamed test files still route correctly.

3. Feature engineering

Daily rows are aggregated to **monthly** buckets per (channel, campaign_type), then enriched with clues built **only from the past** (no leakage):

- **Seasonality:** month, quarter, holiday-season flag (Nov/Dec).
- **Lags:** revenue / spend / ROAS at t-1, t-2, t-3.
- **Rolling:** 3-month trailing mean of revenue / spend / ROAS.
- **Spend / log(spend):** current-period spend is an allowed input (a budget decision); **log1p(spend)** lets the model express diminishing returns.
- **Categoricals:** channel and campaign_type one-hot encoded.

4. Model selection

Chosen: three gradient-boosted regressors (**GradientBoostingRegressor**, **loss="quantile"**, **alpha = 0.1 / 0.5 / 0.9**) trained on **log1p(revenue)**.

Rationale: - **Quantile loss** directly produces calibrated P10/P50/P90 — the required probabilistic output — without distributional assumptions. - **Gradient boosting** captures non-linear spend response and seasonality interactions that a linear model would miss. - **Pooled** single model with channel/type features (rather than one model per channel) because the dataset is small (~102 trainable monthly rows); pooling shares signal and reduces variance. - Trees kept **shallow and regularized** (**max_depth=2**, **min_samples_leaf=5**, **subsample=0.8**, 150 estimators) to resist overfitting on small data.

ROAS is **derived** as predicted revenue / planned spend per quantile (spend is known/decided), keeping revenue and ROAS internally consistent. Quantile crossing is removed by sorting $P10 \leq P50 \leq P90$ per row.

5. Forecasting procedure

Recursive monthly roll-forward: build the next month’s feature row from recent actuals + known calendar + assumed spend, predict, then feed the P50 back as the lag for the following month. Months are summed into 30/60/90-day horizons and rolled up to channel and total.

6. Validation (backtest)

Time-based split (train on earlier months, test on the most recent). Metrics at three grains:

Grain	MAPE (P50)	P10–P90 coverage
Total (scored)	~12.8%	~100%
Per channel	~152%	~75%
Per cell (channelxtype)	~84%	~67%

The total/blended forecast — the primary business and scoring metric — is accurate and well-calibrated. Finer grains are inherently noisy (e.g. Bing revenue swings 10x+ month to month); we communicate this honestly as wider ranges rather than overstating precision.

7. AI integration

A grounded LLM layer (OpenAI), separate from the scored pipeline: - compute concrete **facts** (recent revenue/ROAS, per-channel ROAS, the 90-day forecast), - detect **anomalies** with explainable rules (ROAS swings, spend-up/revenue-down, Meta no-revenue), - prompt the LLM to explain drivers and risks **using only those facts**.

A rule-based fallback guarantees the demo works with no key, and any API error degrades gracefully.

8. Assumptions & limitations

- Meta is spend-only (no revenue attribution in the data).
- Aggregate-period, probabilistic forecasts (not daily, not deterministic).
- Existing channel attribution is the source of truth.
- Horizon quantiles are summed across months (a practical approximation of the horizon interval).
- Small data caps fine-grained accuracy; the design prioritizes a calibrated total forecast plus honest uncertainty.